

HACKATHON

BAI 2026

PROJET Agence OnStream

.IPSSI'



Équipe 25

I.

Société Agence Onstream

Présentation générale du projet

I. Présentation générale du projet

L'agence Onstream est une entreprise spécialisée dans la mesure d'audience publicitaire.

L'objectif de notre mission est de garantir la sécurité des données et le fonctionnement permanent du réseau informatique.

Assurer une continuité de service

Protéger les échanges internes

II.

Choix de la solution technologique

II. Choix de la solution technologique

OPNsense

Pfsense

Notre solution est..

Accessibilité et vigilance

Simplicité mais rigueur

Ergonomie et expertise

II. Choix de la solution technologique

OPNsense

Pfsense

Notre solution est..

Expérience et exigence

Reconnaissance et complexité

Héritage et contraintes

II. Choix de la solution technologique

OPNsense

Pfsense

Notre solution...

OPNsense

Pérennité et Performance

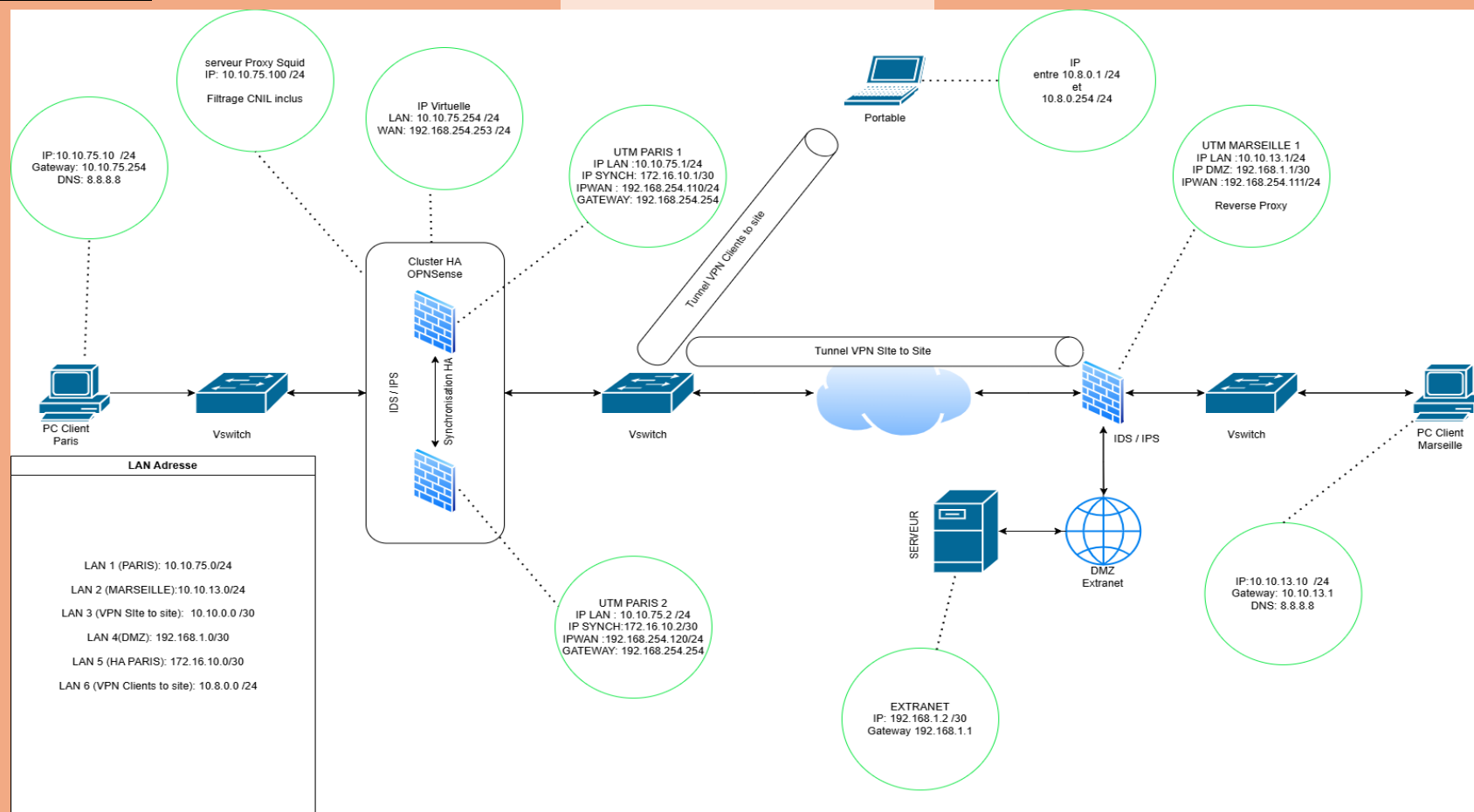


Architecture réseau et topologie

Équipe 25

III. Architecture réseau et topologie

Architecture réseau



III.1

Équipe 25

III.2

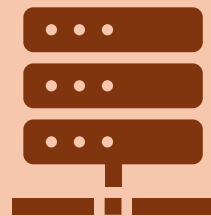
Equipement ou Service	Reseau ou Interface	Adresse IP	Masque (CIDR)	Passerelle (Gateway)
UTM Paris 1 (Actif)	LAN Paris	10.10.75.1	/24	192.168.254.254
UTM Paris 1 (Actif)	Synchronisation HA	172.16.10.1	/30	n/a
UTM Paris 1 (Actif)	WAN	192.168.254.110	/24	192.168.254.254
UTM Paris 2 (Passif)	LAN Paris	10.10.75.2	/24	192.168.254.254
UTM Paris 2 (Passif)	Synchronisation HA	172.16.10.2	/30	n/a
UTM Paris 2 (Passif)	WAN	192.168.254.120	/24	192.168.254.254
IP Virtuelle (VIP)	Cluster Paris	10.10.75.254	/24	n/a
PC Client Paris	LAN Paris	10.10.75.10	/24	10.10.75.254
Serveur Proxy Squid	LAN Paris	10.10.75.100	/24	10.10.75.254
UTM Marseille 1	LAN Marseille	10.10.13.1	/24	n/a
UTM Marseille 1	WAN	192.168.254.111	/24	192.168.254.254
PC Client Marseille	LAN Marseille	10.10.13.10	/24	10.10.13.1
Serveur Extranet	DMZ Marseille	192.169.1.2	/30	192.169.1.1
Tunnel VPN Site a Site	Interconnexion	10.10.0.0	/30	n/a
Pool VPN Clients	Acces Nomades	10.20.0.1 a 10.20.0.254	/24	n/a

IV.

Configuration et mise en place

Équipe 25

LOT 1

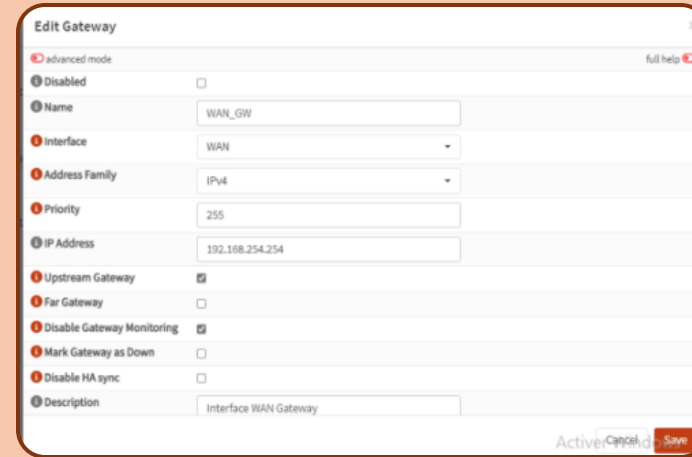
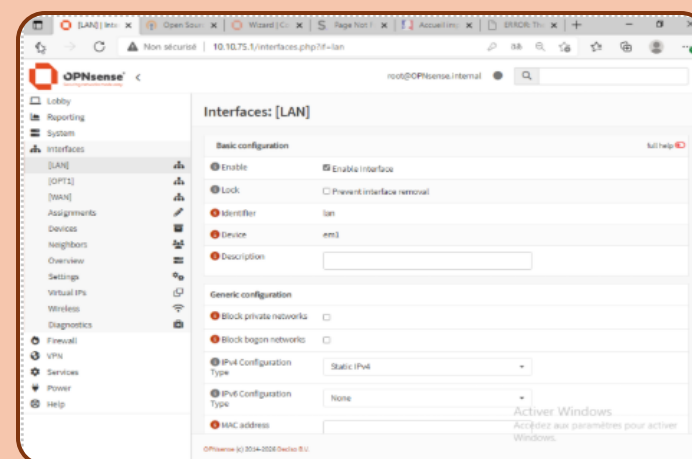
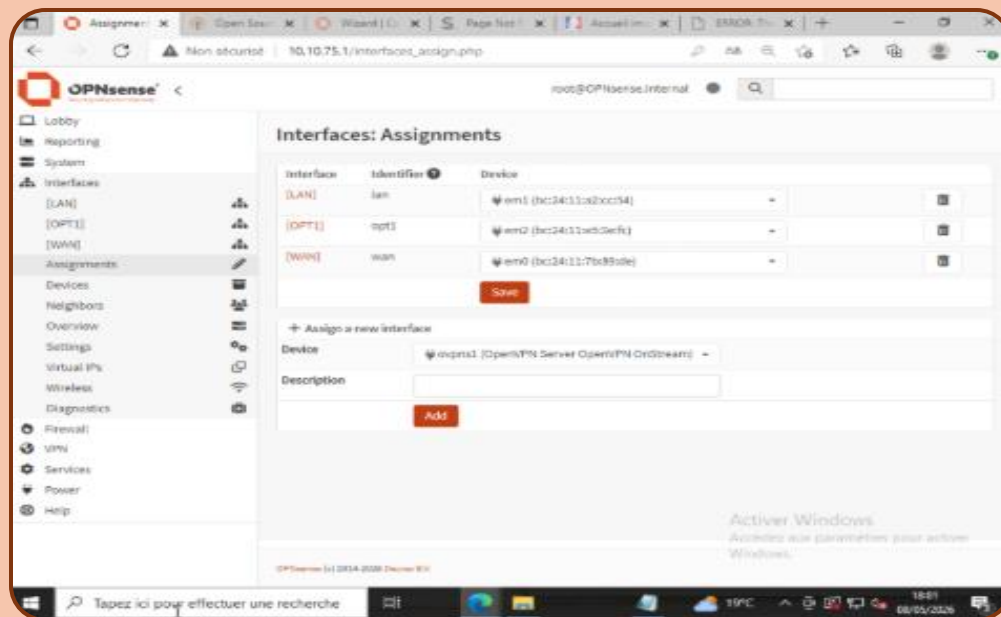


Cluster HA / Firewall Paris et Marseille

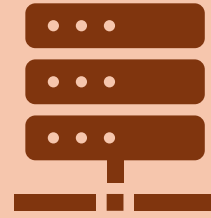
Cluster HA / Firewall Paris et
marseille

VPN inter-sites Paris
- Marseille

Paramétrage des Firewalls pt.1

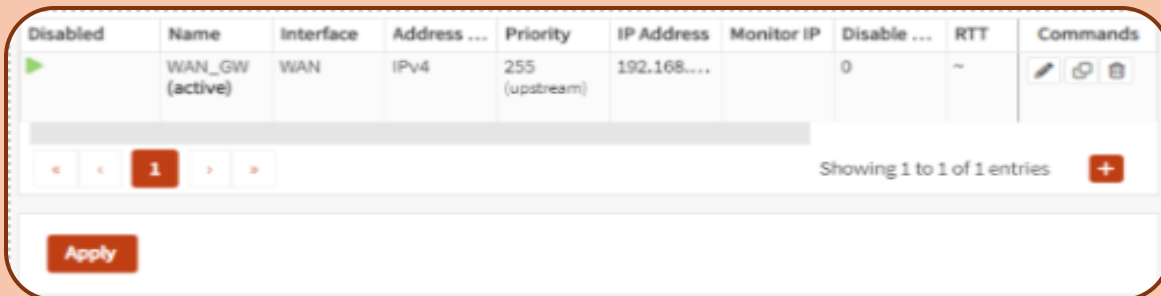
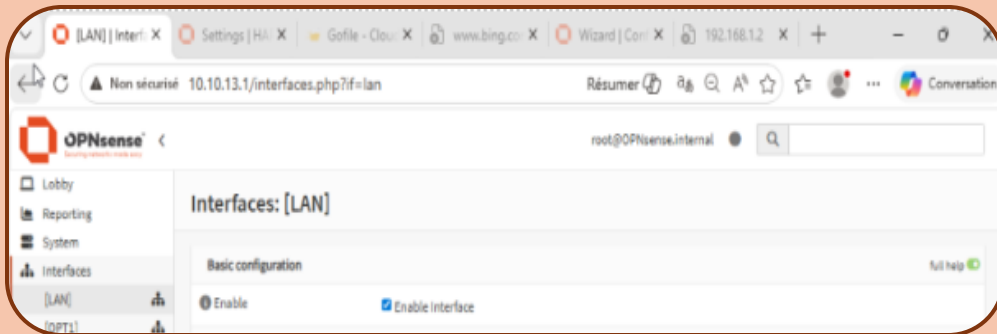


Équipe 25

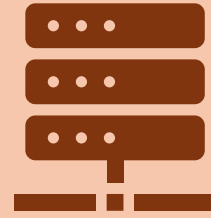


Cluster HA / Firewall Paris et Marseille

Paramétrage des Firewalls pt.2

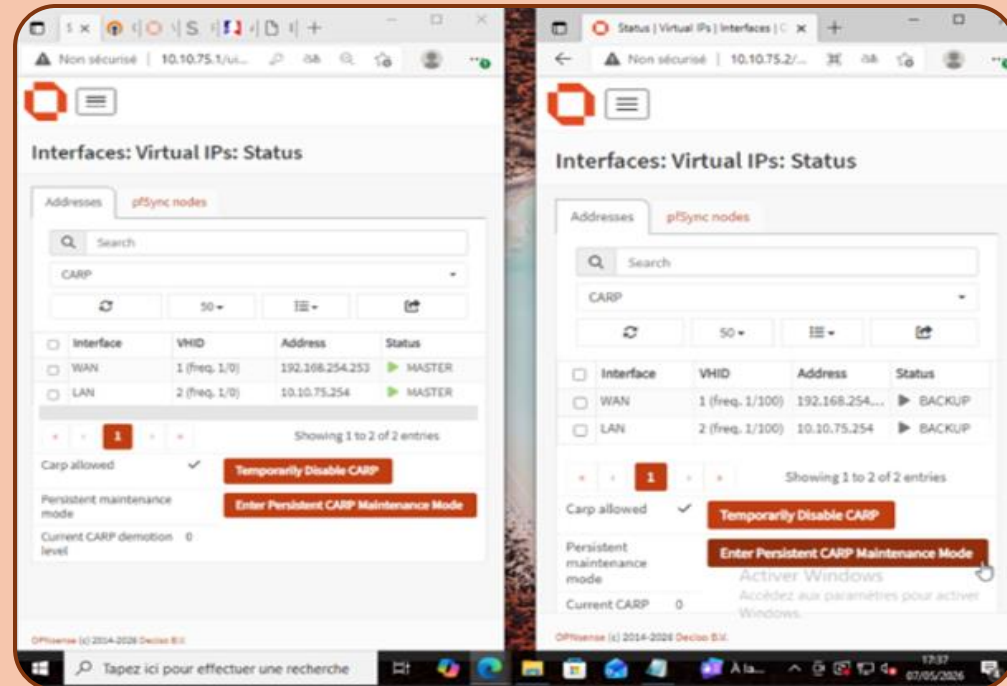
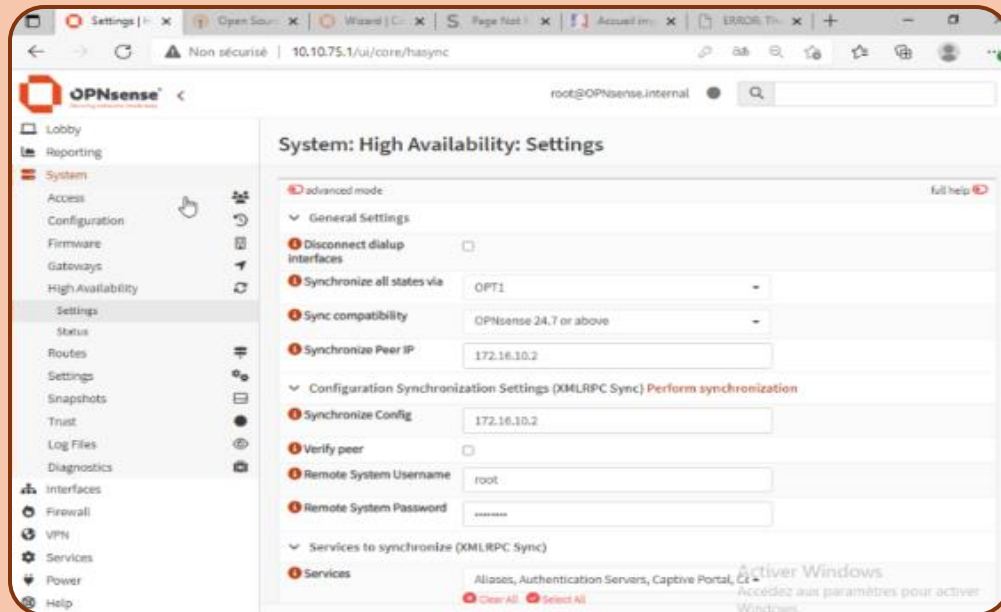


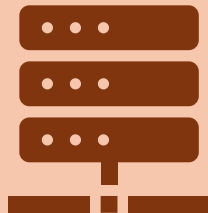
Interface Statistics						
Interface	Bytes In	Bytes Out	Packets In	Packets Out	Errors In	Errors Out
Collisions						
WAN	113.14 MB	16.08 MB	273562	175368	4927	0
0						
OPT1	75.03 MB	80.11 MB	353116	374739	91	0
0						
LAN	5.95 MB	84.69 MB	38433	149317	28226	0
0						



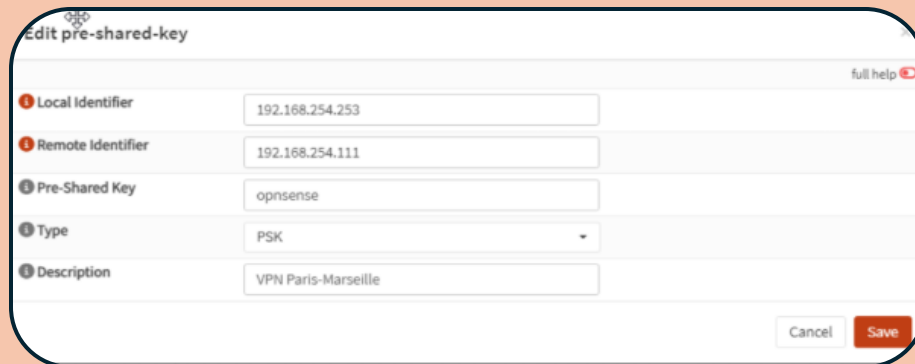
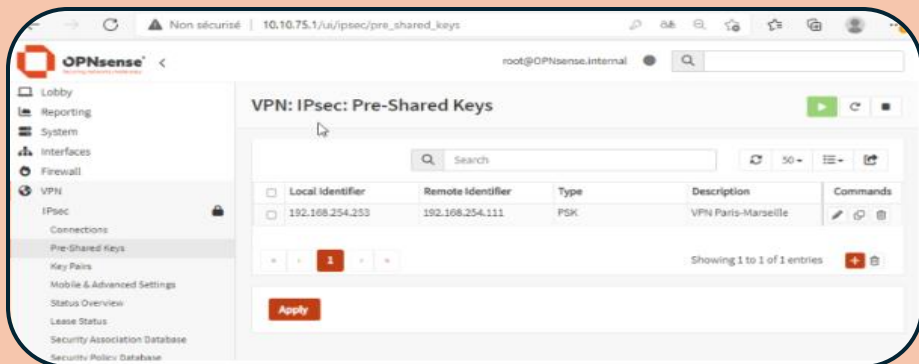
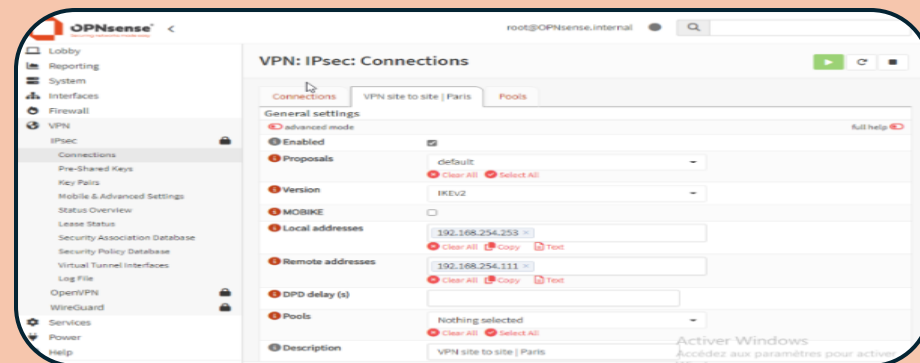
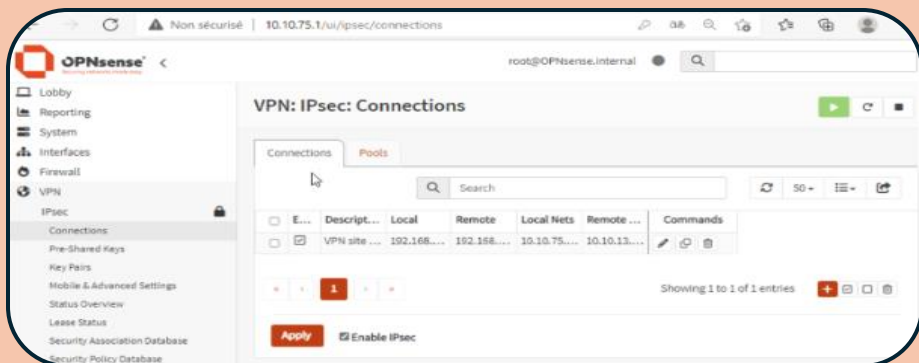
Cluster HA / Firewall Paris et Marseille

Paramétrage du HA





Paramétrage des VPN inter-sites Paris - Marseille





LOT 2

Configuration



Client Paris



Proxy Squid + filtrage
catégoriel



Proxy transparent

Note :

Dépendance : Dépend du FW Paris configuré en Lot 1

VPN Client-to-Site - PKI & Serveur



OpenVPN - Autorité de Certification (CA)

Edit Certificate

Method: Import an existing Certificate Authority

Description: OpenVPN - CA

Output (PEM format)

Certificate data: -----BEGIN CERTIFICATE-----
MIIEUzCCAzugAwIBAgIBADANBgkqhkiG9w0BAQsF
ADCBnDELMakGA1UEBhMCRlIx
FjAUBgNVBAGMDUlsZS1kZS1GcmFuY2UxDjAMBgN
VBACMBVBhcmIzMRgwFgYDVQQLK
-----END CERTIFICATE-----

Private key data: -----BEGIN PRIVATE KEY-----
MIIEvgIBADANBgkqhkiG9w0BAQEFAASCBAkwggS
kAgEAAoIBAQDacZQkwuC6FT2z
Xgb1u2KUiwGOMNDS0rDGO5mnjS/besemS3vUJb
WQzyDZ5JjypV3uudBu0ASLedKr
-----END PRIVATE KEY-----

Serial for next certificate:

Buttons: Cancel, Save

Edit Certificate

Method: Create an internal Certificate

Description: CERT_OpenVPN_Paris

Key

Type: Server Certificate

Private key location: Save on this firewall

Key type: RSA-2048

Digest Algorithm: SHA256

Issuer: OpenVPN - CA

Lifetime (days): 397

Certificat Serveur
OpenVPN

Edit Certificate

Method: Create an internal Certificate

Description: CERT_vpnuser

Key

Type: Client Certificate

Private key location: Save on this firewall

Key type: RSA-2048

Digest Algorithm: SHA256

Issuer: OpenVPN - CA

Lifetime (days): 397

Création utilisateur &
Certificat client

VPN Client-to-Site - Configuration Serveur & Firewall



General Settings

Role: Server

Description: OpenVPN OnStream

Enabled: ☒

Protocol: UDP

Port number: 1194

Bind address:

Type: TUN

Server (IPv4): 10.8.0.0/24

Edit Instance

Auth Token Lifetime:

Routing

Local Network: 10.10.75.0/24

Remote Network:

Interface: WAN

Direction: in

TCP/IP Version: IPv4

Protocol: UDP

Source / Invert: ☐ Use this option to invert the sense of the match.

Source: any

Advanced

Destination / Invert: ☐ Use this option to invert the sense of the match.

Destination: WAN address

Destination port range: from: OPENVPN (1194) to: OPENVPN (1194)

Interface: OpenVPN

Direction: in

TCP/IP Version: IPv4

Protocol: any

Source / Invert: ☐ Use this option to invert the sense of the match.

Source: OpenVPN network

Advanced

Destination / Invert: ☐ Use this option to invert the sense of the match.

Destination: Single host or Network

Destination port range: from: to:

Autorisation règle WAN
UDP port 1194

Autorisation règle
OpenVPN

OpenVPN – Configuration
serveur

Mise en service & Tests

VPN: OpenVPN: Client Export

Remote Access Server OpenVPN OnStream udp:1194

Export type File Only

Hostname 192.168.254.110

Port 1194

Use random local port ☒

Accounts / certificates

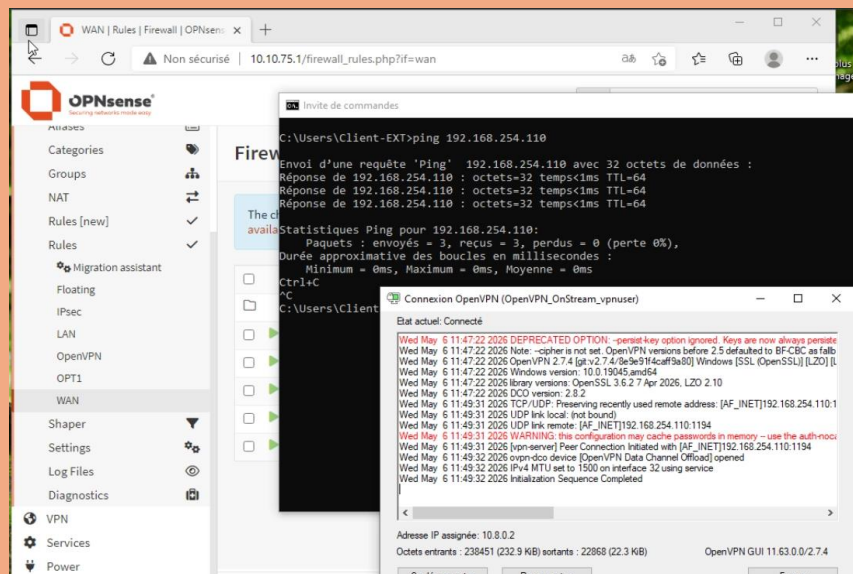
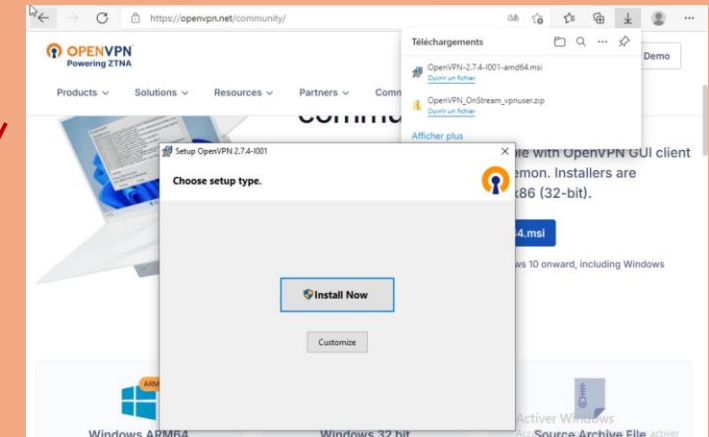
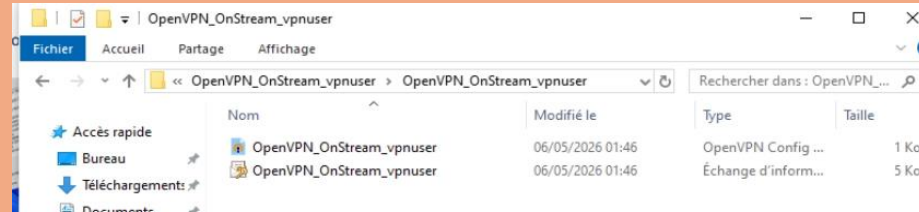
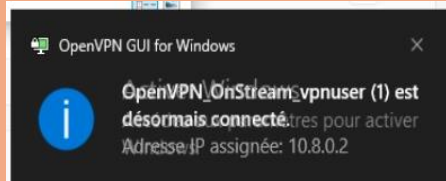
Certificate Linked user

(none) Exclude certificate from export

CERT_OpenVPN_Paris

CERT_vpnuser

Activer Windows
Accédez aux paramètres pour activer



VPN: OpenVPN: Connection Status

Sessions Routes

Search

Filter type

50

Type	Description	Commo...	Username	Real Address	Virtual IPv4 ...	Virtual
server	OpenVPN OnStream	vpnuser	vpnuser	192.168.254.105:65524	10.8.0.2	

Proxy Squid Configuration & filtrage



Administration | Squid Web Proxy | Wizard | Configuration | System | 10.10.75.1/ui/proxy#

root@OPNsense.internal

Edit blacklist

☒ enabled

Filename: Blacklists

URL: https://dsi.ut-capitole.fr/blacklists/download/bla...

username (optional):

password (optional):

categories (if available): adult, agressif, ai, arjel, bank, bitcoin, drogue

ssl ignore cert: ☒

Description: Blacklist url

Cancel Save

10.10.75.1/firewall_rules.php?if=lan

root@OPNsense.internal

Firewall: Rules: LAN

The changes have been applied successfully, remember to update your backup server in System: High availability: status

	Protocol	Source	Port	Destination	Port	Gateway	Schedule	Description	
☐	IPv4 TCP	LAN net	*	127.0.0.1	3128	*	*	Proxy Transparent	
Automatically generated rules									16
☐	IPv4 *	LAN net	*	*	*	*	*	Default allow LAN to any rule	

Services: Squid Web Proxy: Administration

General Proxy Settings | Forward Proxy | Proxy Auto-Config | Remote Access Control Lists | Support

☒ advanced mode

Allowed Subnets: Clear All Copy Text

Unrestricted IP addresses: Clear All Copy Text

Banned host IP addresses: Clear All Copy Text

Whitelist access for banned hosts: ☒

Whitelist: Clear All Copy Text

Blacklist: Test Blacklists httpforever.com https://www.impots.gouv.fr/ Clear All Copy Text

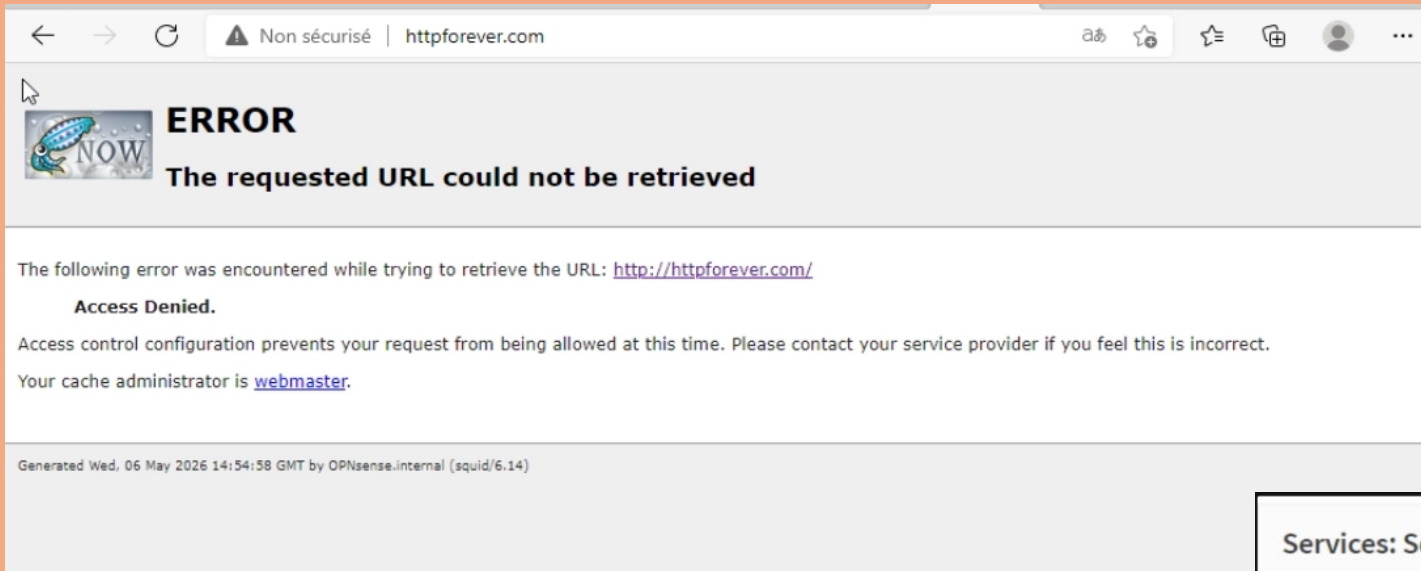
Apply

Activer Windows

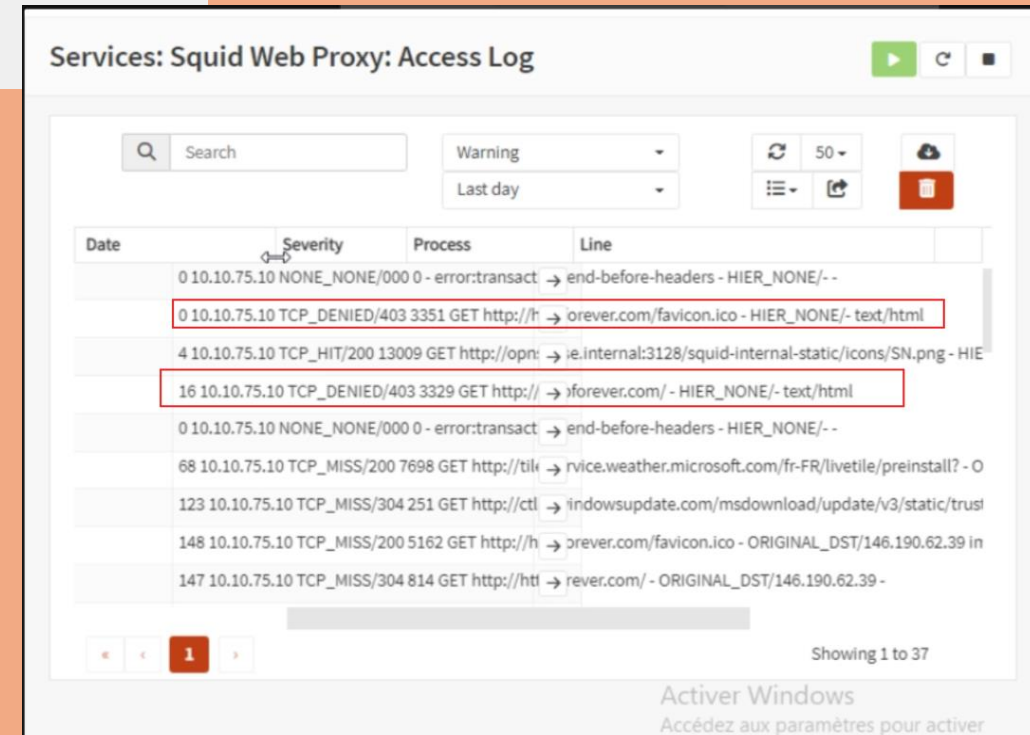
Redirection HTTP
transparente vers Squid
(3128)

Équipe 25

Proxy Squid - Preuve de fonctionnement

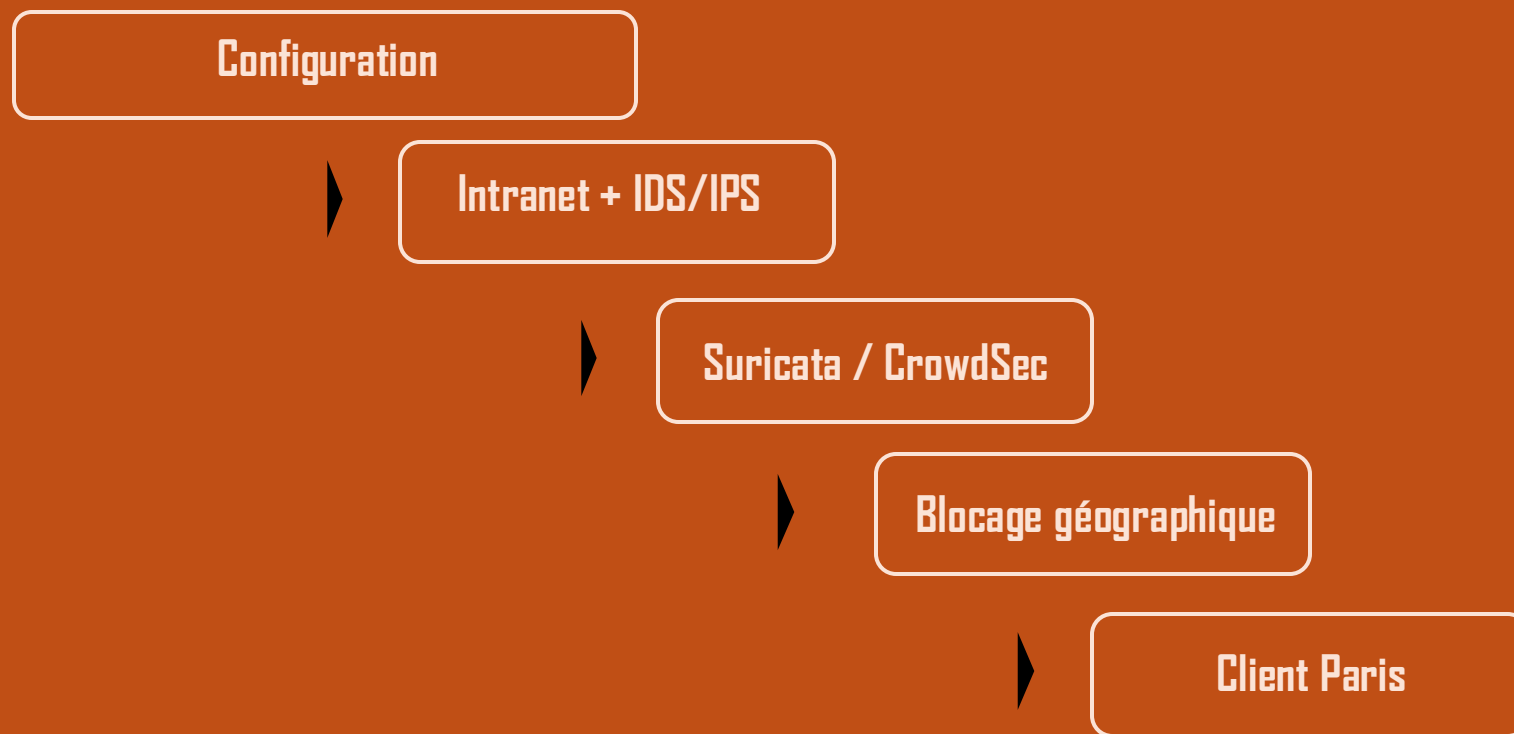


TCP_DENIED/403 →
Requête bloquée par
Squid





LOT 3



Note :

Dépendance : S'appuie sur l'infrastructure Paris des Lots 1 et 2



LOT 3

Configuration du moteur Suricata
en mode IPS Divert

▼ General Settings

Enabled

☒

Capture mode

Divert (IPS)

Listeners

1

▼ Detection

Pattern matcher

Default

▼ Logging

Enable syslog alerts

☒

Enable eve syslog output

☒

Rotate log

Weekly

Save logs

4

Apply

← →

100%

+

⌵

Active Windows



SURICATA®

Équipe 25



LOT 3

Activation des règles de blocage et de surveillance.

☐	sid	Action	Source	ClassType	Message	Info / Enabled
☐	2000005	drop	emerging-exploit.rules	attempted-dos	ET EXPLOIT Cisco Telnet ...	☐ ☒
☐	2000007	drop	emerging-exploit.rules	attempted-dos	ET EXPLOIT Catalyst SSH ...	☐ ☒
☐	2000031	drop	emerging-exploit.rules	attempted-admin	ET EXPLOIT CVS server he...	☐ ☒
☐	2000048	drop	emerging-exploit.rules	attempted-admin	ET EXPLOIT CVS server he...	☐ ☒
☐	2000049	drop	emerging-exploit.rules	attempted-admin	ET EXPLOIT CVS server he...	☐ ☒
☐	2000342	drop	emerging-exploit.rules	misc-attack	ET EXPLOIT Squid NTLM ...	☐ ☒
☐	2000372	drop	emerging-exploit.rules	attempted-user	ET EXPLOIT MS-SQL SQL I...	☐ ☒
☐	2000373	drop	emerging-exploit.rules	attempted-user	ET EXPLOIT MS-SQL SQL I...	☐ ☒
☐	2000377	drop	emerging-exploit.rules	attempted-admin	ET EXPLOIT MS-SQL heap...	☐ ☒
☐	2000378	drop	emerging-exploit.rules	attempted-dos	ET EXPLOIT MS-SQL DOS ...	☐ ☒
☐	2000379	drop	emerging-exploit.rules	attempted-dos	ET EXPLOIT MS-SQL DOS ...	☐ ☒
☐	2000380	drop	emerging-exploit.rules	attempted-admin	ET EXPLOIT MS-SQL Spik...	☐ ☒
☐	2000488	drop	emerging-exploit.rules	attempted-user	ET EXPLOIT MS-SQL SQL I...	☐ ☒

☐	sid	Action	Source	ClassType	Message	Info / Enabled
☐	2000006	drop	emerging-do...	attempted-dos	ET DOS Cisco...	☐ ☒
☐	2000010	drop	emerging-do...	attempted-dos	ET DOS Cisco...	☐ ☒
☐	2000011	drop	emerging-do...	attempted-dos	ET DOS Catal...	☐ ☒
☐	2000378	drop	emerging-ex...	attempted-dos	ET EXPLOIT ...	☐ ☒
☐	2000379	drop	emerging-ex...	attempted-dos	ET EXPLOIT ...	☐ ☒

☐	sid	Action	Source	ClassType	Message	Info / Enabled
☐	2404300	drop	botcc.rules	trojan-activity	ET CNC Feod...	☐ ☒
☐	2404301	drop	botcc.rules	trojan-activity	ET CNC Feod...	☐ ☒
☐	2404302	drop	botcc.rules	trojan-activity	ET CNC Feod...	☐ ☒
☐	2404303	drop	botcc.rules	trojan-activity	ET CNC Feod...	☐ ☒
☐	2404304	drop	botcc.rules	trojan-activity	ET CNC Feod...	☐ ☒



LOT 3

Firewall: Aliases 1% (51965/9148500)

[Aliases](#) [GeoIP settings](#) [Actions](#)

[full help](#)

Last updated	2026-05-08T15:39:54
Total number of ranges	1200277
Url	https://1342121:ZVDNRu_v354IK62TUjqxeGuHnV...

Configuration de Maxmind sur notre Firewall

GeoLite2 Country: CSV Format

Edition ID: GeoLite2-Country-CSV	Download ZIP Download SHA256 Get Permalinks
Format: GeoIP2 CSV (docs)	
Updated: 2023-01-06	





LOT 3

Création de L'alias + de la règle WAN

Edit Alias

Enabled ☒

Name

Type

Categories

Content

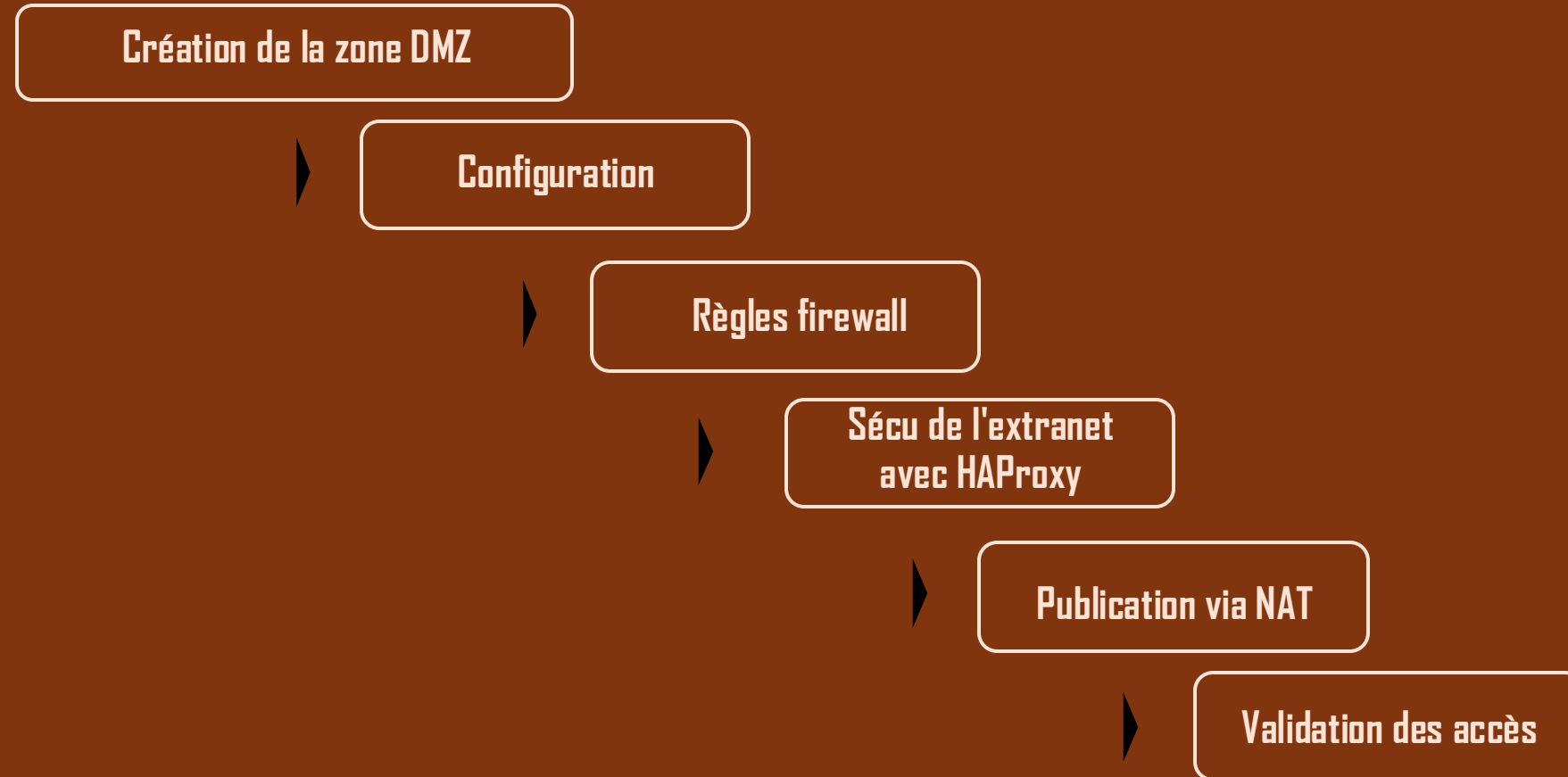
Region	Countries		
Africa	Nothing selected	-	☐
America	Nothing selected	-	☐
Antarctica	Nothing selected	-	☐
Arctic	Nothing selected	-	☐
Asia	☒ China, ☒ Korea (North)	-	☐
	2 out of 49 selected		
Atlantic	Nothing selected	-	☐
Australia	Nothing selected	-	☐
Europe	☒ Russia, ☒ Ukraine	-	☐
	2 out of 50 selected		
Indian	Nothing selected	-	☐
Pacific	Nothing selected	-	☐

Activater Windows
Accédez aux paramètres

	Protocol	Source	Port	Destination	Port	Gateway	Schedule	Description	
								Automatically generated rules	
	IPv4 *	GeolPite2	*	*	*	*	*	GeolIP block location	



LOT 4



Note :

Dépendance : Dépend du FW Marseille configuré en Lot 1

Équipe 25



LOT 4

Création de l'interface DMZ

OPNsense® Securing networks made easy

root@OPNsense.internal

Interfaces: Assignments

Interface	Identifier	Device
[LAN]	lan	em1 (bc:24:11:f1:d0:05)
[OPT1]	opt1	em2 (bc:24:11:89:36:c5)
[WAN]	wan	em0 (bc:24:11:49:7c:22)

Save

Interface dédiée à la zone DMZ

OPNsense® Securing networks made easy

root@OPNsense.internal

Enable ☒ Enable Interface

Lock ☐ Prevent interface removal

Identifier opt1
The internal configuration identifier of this interface.

Device em2

Activation de l'interface dédié à la DMZ



LOT 4

Configuration Interface

OPNsense[®] Securing networks made easy < root@OPNsense.internal

Lobby

Reporting

System

Interfaces

[LAN]

[OPT1]

Logons are prefixes that should never appear in the internet routing table, address in any packets you receive.

IPv4 Configuration Type	Static IPv4
IPv6 Configuration Type	None
MAC address	

Static IPv4 configuration

IPv4 address	192.168.1.1	30
--------------	-------------	----

Adresse IP de la
passerelle DMZ

Équipe 25



LOT 4

Isolation de la DMZ avec les règles Firewall

Firewall: Rules: OPT1

Select category

	Protocol	Source	Port	Destination	Port	Gateway	Schedule	Description
☐	IPv4 *	192.168.1.0/30	*	*	*	*	*	Automatically generated rule
☐	IPv4 *	192.168.1.0/30	*	10.10.13.0/24	*	*	*	Regle d'isolation de la DMZ
☐	IPv4 *	192.168.1.0/30	*	10.10.75.0/24	*	*	*	Isolation DMZ vers Paris

pass
pass (disal)

Destination

Single host or Network

10.10.13.0/24

Destination

Single host or Network

10.10.75.0/24

Firewall: Rules: OPT1

Edit Firewall rule

Action

Block

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is not delivered to the destination.

Disabled

☒ Disable this rule

Set this option to disable this rule without removing it from the list.

Quick

☒ Apply the action immediately on match.

If a packet matches a rule specifying quick, then that rule is considered the last matching rule and the action is taken. When a rule does not have quick enabled, the last matching rule wins.

Règles
d'isolation

.IPSSI

Réseaux internes de
l'entreprise

Blocage des accès vers le
LAN

Équipe 25



LOT 4

Sécurisation de l'extranet avec HAProxy

OPNsense
Securing networks made easy

- Reporting
- System
- Interfaces
- Firewall
- VPN
- Services
 - Captive Portal
 - DHCRelay
 - Dnsmasq DNS & DHCP
 - HAProxy

Services: HAProxy: Settings

Introduction Real Servers Virtual Services Rules & Checks User Management Settings Advanced

☐	Enabled	Server Name	Type	Mode	Server Address	Server Port	Description
☐	☒	Web-Marseille	static	active [default]	192.168.1.2	80	

Introduction Real Servers Virtual Services Rules & Checks

☐	Enabled	Rule Name
☐	☒	Block_post

Introduction Real Servers Edit Condition

☐	Condition Name	
☐	is_post	
Name		is_post
Description		
Condition		

Services: HAProxy: Settings

Introduction Real Servers Virtual Services

☐	Enabled	Backend Pool Name
☐	☒	Backend_extranet

Introduction Real Servers Virtual Services

☐	Enabled	Public Service Name
☐	☒	Front_extranet

Server web réel
en DMZ

Blocage des
requêtes POST
pour sécuriser
.IPSSI

Équipe 25

Point d'entrée public



LOT 4

Publication de l'extranet via NAT

OPNsense Securing networks made easy

root@OPNsense.internal

Firewall: NAT: Destination NAT

Categories Tree Search

			Version	Protocol	Source	Port	Destination	Port	Redirect Ta...
☐	☒	LAN	*	TCP	*	*	LAN address	80	*
☐	☒	WAN	*	TCP	*	*	192.168.254.11/24	http	192.168.1.2/24

Source NAT
Destination NAT

Edit Rule

Categories Nothing selected
Clear All Select All
Assign categories for rule organization.

Description WAN to extranet DMZ
Enter a description to identify this rule.

Interface WAN
Clear All Select All
Choose the interface(s) on which the traffic originates.

Destination address or alias to match.

Destination Port HTTP (80)
Destination port or port range.

Translation

Redirect Target IP Single host or Network
192.168.1.2/24
The internal IP address to forward traffic to.

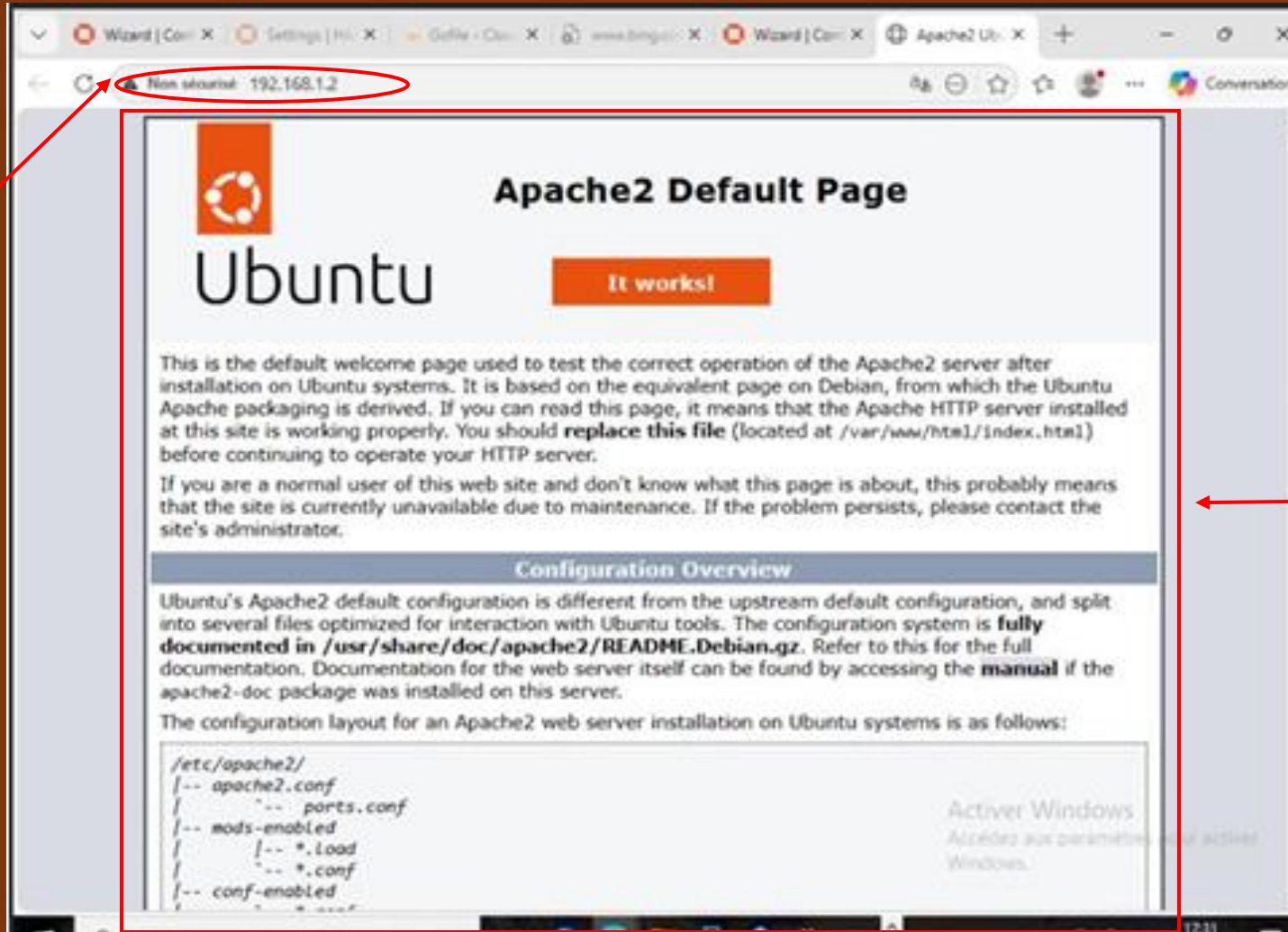
Équipe 25



LOT 4

Validation de l'accès au service

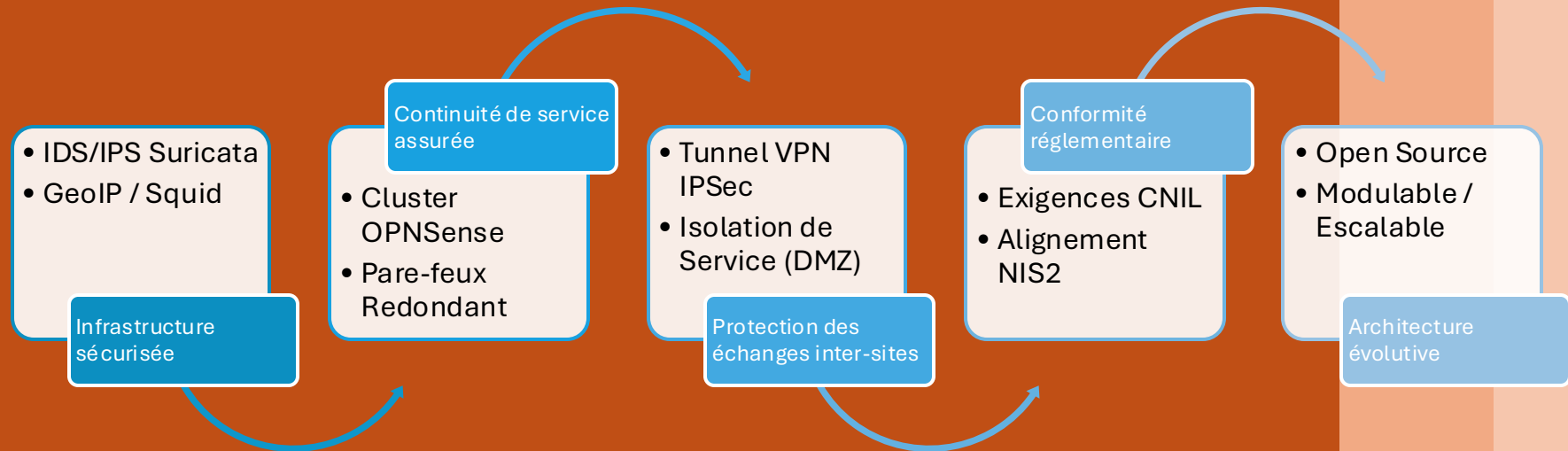
IP publique



Service accessible depuis Internet

V.

CONCLUSION



VI.

Proposition Financiere

Une solution sécurisée avec un coût maîtrisé

Prestations de services	Unite	Qte	P.U. HT	Total HT
Conception du Document d'Architecture Technique (DAT)	Jour(s)	0.5	750,00 €	375,00 €
Lot 1 - Deploiement cluster Firewall Paris HA (CARP/pfsync), VPN IPSec inter-sites et VPN Client-to-site OpenVPN	Jour(s)	4	750,00 €	3 000,00 €
Lot 2 - Deploiement proxy transparent Squid, filtrage categoriel SquidGuard et journalisation CNIL	Jour(s)	1.5	750,00 €	1 125,00 €
Lot 3 - Deploiement IDS/IPS (Suricata/CrowdSec) et blocage geographique (RU, UA, CN, KP)	Jour(s)	1.5	750,00 €	1 125,00 €
Lot 4 - Deploiement Firewall Marseille standalone, Extranet DMZ, reverse proxy HAProxy et serveur web LXC	Jour(s)	1.5	750,00 €	1 125,00 €
Pilotage de projet et recette globale de l'interconnexion des lots	Jour(s)	0.5	750,00 €	375,00 €
Transfert de Compétences / Formation Administrateur	Jour(s)	0.5	750,00 €	375,00 €
Sous-total prestations HT				7 500,00 €

Total HT	9 905,00 €
TVA 20 %	1 981,00 €
Total TTC	11 886,00 €

Fourniture de materiel	Unite	Qte	P.U. HT	Total HT
Appliance firewall Protectli VP4630 - 6x Intel NIC 2.5G, Intel Core i3-8145U, 8 Go RAM, 120 Go SSD (x2 cluster HA site Paris)	U.	2	695,00 €	1 390,00 €
Appliance firewall Protectli VP4630 - 6x Intel NIC 2.5G, Intel Core i3-8145U, 8 Go RAM, 120 Go SSD (site Marseille)	U.	1	695,00 €	695,00 €
Forfait cablage et accessoires reseau (patch panels, cables Cat6A, organiseurs de cables)	Forfait	1	320,00 €	320,00 €
Sous-total materiel HT				2 405,00 €

Équipe 25

VII.

Plan de charge

